

Inscrutability or Trilemma? Methodological Disagreement in Empirical Climate-Economy Research and Its Policy Implications

Charley Baker, Ronnie Obeng-Osei, Phyllis Weinheimer

Introduction

Translating climate damages into quantifiable economic values is indispensable in assessing climate economics and yet the social cost of carbon, which does exactly that, is highly contested. As the Pigouvian price designed to internalize the externality of greenhouse gas emissions (GHG), it anchors regulatory ambition in countries like the United States, feeds into the financial risk assessments that central banks and the NGFS conduct for climate stress tests, and underpins the cost-benefit logic of major decarbonization policies. The reliability of the damage functions that produce it therefore have implications beyond the academic sphere. Recently published (April 2026) papers raise urgent and competing doubts. Firstly, Lemoine, Hausman, and Shrader (hereafter LHS) characterize the climate damage estimation as a "problem from hell," navigable through methodological trade-offs but not definitively solvable, while Curtin and Burgess (hereafter CB) go further, arguing that the macro-level climate-economy relationship is "empirically inscrutable," and that aggregate damage estimates lack coherent causal interpretation.

This essay argues that the disagreement between LHS and CB reflects different standards rather than fundamentally different views of the empirical record with both offering important insights. Top-down, country-level panel regressions of GDP on temperature face identification problems in line with CB's argument, but the methodological landscape is not uniformly bleak and sectoral and structural approaches do indeed provide partial but useful policy evidence as LHS conclude. What policymakers should take from this debate is a reorientation of climate damage numbers away from aggregate damage functions and single point estimates, toward credibly identified sectoral studies.

Lemoine, Hausman, and Shrader's Methodological Trilemma

The central contribution of LHS' paper is to impose order on a fragmented methodological debate through the climate impacts trilemma. Any method for estimating climate damages can achieve at most two of the three criteria. The first requirement is robustness to economic model structure, where results do not depend critically on specific assumptions about how production, adaptation, or international trade operate. The second is its interpretability as effects of persistent, widespread, anticipated climate change, as opposed to short-lived weather anomalies that are transitory. The third is quasi-experimental econometric identification, which requires the variation exploited by the researcher to be as-good-as-randomly assigned relative to potential confounders.

Each major methodological tradition in the literature sacrifices a different element of the trilemma. Panel regressions, which identify the effect of year-to-year temperature variation within countries using fixed effects, provide the cleanest causal identification currently available. But the variation they exploit is

precisely the kind that individuals do not fully anticipate and therefore do not fully adapt to. LHS note that panel methods capture only short-run myopic responses and miss how capital stocks, trade patterns, and forward-looking investment decisions adjust to persistent climate change. Cross-sectional approaches, which compare economies operating at systematically different mean temperatures, can capture long-run equilibrium adaptation but suffer from omitted variable bias and endogeneity. Factors determining a country's climate also shape its productivity, governance, and economic history. Structural models can in principle capture both persistence and identification, but their results are only as credible as the modelling and parametric assumptions imposed.

LHS do not treat this trilemma as grounds for abandoning estimation and their argument is that methodological diversity is informative in its own right. When panel and structural approaches converge on similar damage magnitudes, that convergence provides methodological support despite neither method being individually sufficient. When they diverge, this identifies where adaptation dynamics or general equilibrium effects are doing the most analytical work, pointing future research toward the places where the trilemma binds hardest. Hybrid strategies that use one method to test the assumptions of another represent the frontier of the field.

The LHS treatment of adaptation is the paper's most important analytical contribution, and its sharpest departure from conventional practice. Panel regressions are typically interpreted as capturing some blend of short- and long-run responses, but LHS show that what they identify is much closer to the myopic end of the spectrum. Panel methods identify only the unanticipated component of temperature variation, and therefore systematically miss adaptation that has already occurred. But policymakers need damage estimates of persistent climate change with adaptation and not just the surprise temperature shock. Since the methods are built around that unanticipated component, no additional data coverage can overcome this.

Curtin and Burgess' Argument for Irreducible Inscrutability

CB's critique is at once narrower in target and more radical in implication than LHS's. Narrower, due to its focus on top-down country-year panel regressions of GDP on temperature rather than the full spectrum of methods LHS survey. Radical, because it argues that the aggregate climate-economy relationship cannot be recovered from macro data not due to limitations of identification or data, but because no stable aggregate causal relationship of the relevant kind exists.

The core of CB's argument is a degrees-of-freedom problem that runs deeper than concerns about statistical power. The climate-economy relationship varies across countries for reasons that are not noise but signal. For example, El Salvador and Iraq are both hot but their productive structures, institutions, and adaptive capacities differ profoundly. The relationship also varies temporally within countries as development changes the composition of output and the capacity for adaptation. For instance, India in 1970 and 2020 are climatically similar but economically transformed. To estimate a pooled relationship from country-year data, researchers must either allow this variation to enter the model or assume it away. Allowing it produces a model too parameterized to estimate credibly, whereas assuming it away collapses

variation across countries and time into a single coefficient that corresponds to no specific place or period. The resulting estimate is a statistical summary whose causal interpretation is, at best, indeterminate.

CB's empirical replication work lends particular force to this theoretical critique. The authors demonstrate that prominent results in the literature, including those of Burke, Hsiang, and Miguel (2015), are highly sensitive to a small number of influential observations. Events such as the Rwandan genocide in 1994 happened to coincide with temperature anomalies, and removing fewer than ten observations can shift estimated damage coefficients by between 25 and 33 percent. When these climate-unrelated institutional events coincide with weather shocks, the regression absorbs both into a single coefficient, and what is identified becomes a correlation between political disaster and weather rather than the causal effect of temperature on economic productivity.

The retraction of Kotz, Levermann, and Wenz (2024) validates this concern in a highly visible case. The authors initially produced the headline finding that climate damages would outstrip mitigation costs sixfold, but this was retracted in late 2025 after a dataset error was identified that fundamentally altered the result. The retraction does not establish that every aggregate estimate is wrong, but it illustrates precisely the specification fragility CB diagnose as a structural feature of top-down estimation. CB's framework predicts this kind of sensitivity not as aberration but as the expected consequence of trying to identify a weak climate signal through highly volatile macro data.

Comparison between the Papers

Before comparing the two papers, it is essential to recognize that LHS and CB are, to a substantial degree, critiquing different literatures. LHS survey the full methodological spectrum, from time-series to structural models, and their trilemma applies across all approaches. CB focus on top-down country-year panel regressions. These regressions matter because they are the backbone of the aggregate damage functions that feed into integrated assessment models and social cost of carbon calculations, used in policymaking. But CB's critique does not straightforwardly extend to sectoral studies that operate at finer geographic scales and with more direct causal mechanisms.

The sharpest conceptual disagreement between the two papers concerns whether the identification problems they both recognize amount to a problem of external validity or a deeper problem of causal coherence. LHS frame the core limitation of panel methods as one of external validity. Weather shocks are not climate change, and responses to the former may not predict responses to the latter, though addressable through better (hybrid) methods. CB's framing is more demanding, again disputing that the aggregate causal object, the effect of a persistent shift in mean temperature on aggregate GDP across heterogeneous countries over time, does not exist in a form that a pooled regression can recover. External validity presupposes a valid internal identification target where CB deny that the internal target is coherent.

Adaptation is at the centre of this disagreement, and the papers' stances map onto their broader epistemological positions in revealing ways. LHS treat adaptation as an identification challenge with a research programme around it. Hybrid methods that combine panel estimates of short-run responses with cross-sectional comparisons of long-run equilibria can partially reconstruct what durable adaptation

achieves. The structural gravity approach of Astier et al. (2026) illustrates one productive extension of this agenda. By modelling how climate change propagates through international trade and comparative advantage, the authors capture a general equilibrium channel that conventional panel estimates miss. This comes at the cost of model robustness, since the results depend on a specific trade model, but it contributes evidence about a damage channel that LHS identify as systematically underestimated by reduced-form panel methods.

On the other hand, CB treat adaptation not as something that better methods could account for, but as a reason why the underlying relationship itself changes over time. The locally optimal production environment for any given country is unobservable. Researchers cannot determine from aggregate data how much of a country's current economic structure reflects adaptation to its climate rather than historical accident, institutional path dependence, or development trajectory. This means researchers cannot separate the causal effect of temperature from the accumulated adaptive responses to temperature that are already embedded in observed outcomes. Kahn's (2026) distinction between real-time information shocks and ex-post statistical shocks reinforces this point. The adaptation that has already occurred in response to anticipated trends is precisely what panel methods are blind to.

Burke, Zahid, Diffenbaugh, and Hsiang (2026) offer a useful point of comparison in this debate. Their framework translates country-level emissions into estimates of global economic damages using climate projections and an empirically estimated temperature-GDP relationship. This is the kind of top-down aggregate approach CB criticize. Yet one of their central findings speaks directly to CB's concern about temporal heterogeneity. The temperature-GDP relationship has not shifted meaningfully since 1960, despite per-capita incomes roughly tripling over the same period (Burke et al. 2026, Extended Data Fig. 2). If economic development were progressively reducing the climate sensitivity of aggregate output, as CB's temporal heterogeneity argument implies it should, this 60-year stability would be difficult to explain. The result is not a definitive rebuttal of CB's theoretical critique, but it raises the question of why the instability they predict is not more evident in the data..

Overall, LHS' trilemma is the more durable analytical contribution of the two papers, and likely to define how methodological debates in climate economics are framed for the next decade. Its strength is not in any particular empirical finding but in providing a vocabulary for scrupulous diagnosis. Researchers can use it to characterize their own approach transparently, to identify which criteria have been sacrificed, and to seek complementary evidence that addresses what their primary method misses. This is the kind of framework that improves literature over time and is a more valuable contribution than yet another damage estimate.

LHS's constructive stance is also, however, where the paper most invites challenge. The paper is relatively optimistic that existing approaches can still provide useful guidance for policy, particularly when different methods produce similar answers. This optimism sits uneasily with the degree of specification sensitivity that CB document empirically. If removing fewer than ten observations can shift headline damage estimates by a quarter to a third, then convergence across methods needs to be assessed with much more care than LHS provide. The concern is not that convergence across methods is uninformative. It is

whether the apparent agreement across methods is robust enough to justify the confidence the paper places in it.

CB's empirical contribution is important and underappreciated, but the paper risks being read as a broader dismissal than its argument warrants. The degrees-of-freedom critique and the influential-observations finding are significant and apply forcefully to top-down GDP regressions. They apply much less clearly to sectoral studies with tighter causal designs. Carleton et al. (2022), for instance, estimate global mortality damages from climate change using granular variation in mortality rates across temperature bins, with individual-level data and a biological dose-response mechanism. This kind of study does not pool qualitatively heterogeneous economic relationships across dozens of countries in the way CB critique. Their critique is persuasive for aggregate GDP regressions, but it is not obvious that it applies with the same force to studies built on more granular sources of variation. Estimates based on well-identified channels, such as temperature-related mortality, may remain policy-relevant even if the aggregate climate-GDP relationship is difficult to interpret.

Policymaking Discussion and Conclusion

A policymaker reading this literature carefully should, above all, resist the temptation to anchor on a single preferred number. Burke et al. (2026) estimate that one tonne of CO₂ emitted in 1990 will generate an additional \$1,840 in discounted global damages through 2100, with a 90 per cent confidence interval spanning \$500 to \$5,700. That 11-fold range, generated within a single well-executed study under conservative assumptions, already signals that the level of uncertainty is substantial. Across different methodological approaches and modelling choices, the range widens further. CB are right that using a single top-down aggregate estimate as the anchor for cost-benefit analysis of major policy interventions involves a degree of false precision that the evidence does not support.

Equally, policymakers should not infer from this debate that climate damages are unknowable or that the case for policy action rests on evidence that cannot be sustained. The sectoral literature provides real signals. Mortality studies (Carleton et al. 2022), agricultural productivity analyses (Schlenker and Roberts 2009), and energy demand studies provide credibly identified evidence that sustained temperature increases cause harm across economically significant dimensions. These estimates are not immune to the problems LHS diagnose, particularly around the external validity of short-run weather shock identification for anticipating long-run climate change. But they are substantially more robust to CB's degrees-of-freedom critique, and they converge, across quite different identification strategies, on the direction and approximate magnitude of harm.

For central banks and climate financial risk institutions, the methodological debate implies a redesign of how damage functions are used rather than their abandonment. Climate stress tests that anchor physical risk assessments on a single damage function specification are fragile in precisely the way CB predict. The appropriate approach is to stress-test across a range of specifications, treating the methodological debate as itself a source of quantifiable risk. When diverse approaches, from sectoral bottom-up estimates to aggregate top-down regressions, converge on the sign and rough order of magnitude of harm, that convergence is actionable. No credible model in the literature implies that warming of 2°C or more is

economically neutral or beneficial at the global scale. The uncertainty is about the scale of harm, not its direction. Scenario and counterfactual analysis across a range of estimates address these concerns and remain informative in the policymaking sphere, particularly as climate change becomes a growing risk to public finances. For example, the Office for Budget Responsibility (2025) assesses UK fiscal risk under a range of climate damage scenarios, from a below-2°C pathway to higher-warming and higher-GDP-loss variants, rather than collapsing this into one number.

The deepest implication of LHS and CB together concerns the allocation of research effort rather than any particular empirical result. Both papers, from quite different vantage points, suggest that continuing to produce more aggregate panel regressions from the same country-year data is likely to yield diminishing returns to policy-relevant knowledge. The path toward more useful evidence has several components. One is bottom-up sectoral integration that preserves credible identification while broadening the range of channels captured. Another is the use of hybrid approaches that exploit the complementarities highlighted by the LHS trilemma. A third is transparent communication of uncertainty, as exemplified by Burke et al. (2026), who report results across a range of discount rates, time horizons, and specifications rather than obscuring the extent of what remains unknown. The social cost of carbon will remain uncertain. Communicated honestly, that uncertainty is itself a policy-relevant input, one that favours robust strategies over fragile optima and complementary policy instruments over exclusive reliance on price-based tools calibrated to a number the data cannot pin down. For policymakers, the challenge is therefore not to find certainty where none exists, but to make robust decisions in a world where uncertainty itself is one of the most important facts.

References

- Astier, J., et al. (2026). “Quantifying Climate Damages When Regions Trade: A Structural Gravity Approach.” Working paper.
- Burke, M., Zahid, M., Diffenbaugh, N. S. and Hsiang, S. (2026). “Quantifying climate loss and damage consistent with a social cost of carbon.” *Nature*, 651, 959–966.
- Burke, M., Hsiang, S. M. and Miguel, E. (2015). “Global non-linear effect of temperature on economic production.” *Nature*, 527, 235–239.
- Carleton, T., et al. (2022). “Valuing the global mortality consequences of climate change accounting for adaptation costs and benefits.” *Quarterly Journal of Economics*, 137(4), 2037–2105.
- Curtin, J. and Burgess, R. (2026). “The empirically inscrutable climate-economy relationship.” SocArXiv preprint, April 2026.
- Kotz, M., Levermann, A. and Wenz, L. (2024). Retracted. *Nature*, 2024 (retracted December 2025).
- Lemoine, D., Hausman, C. and Shrader, J. (2026). “Navigating the ‘Problem from Hell’: A Guide to Climate Damages.” NBER Working Paper 34348.
- Office for Budget Responsibility (2025). “Fiscal risks and sustainability – July 2025.” London: Office for Budget Responsibility. Available at: <https://obr.uk/frs/fiscal-risks-and-sustainability-july-2025/> (Accessed: 22 June 2026).
- Schlenker, W. and Roberts, M. J. (2009). “Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change.” *Proceedings of the National Academy of Sciences*, 106(37), 15594–15598.